International Islamic University Chittagong

Department of Computer Science and Engineering
B. Sc. in CSE
Mid Exam, Spring-2026

Course Code: CSE 3631 **Course Title**: Operating System

Time: 1 hours 30 minutes **Full Marks**: 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes are mentioned in additional Columns

[Answer the questions from the followings]

1. a) List four services provided by operating system and explain how each creates convenience for users.Explain the difference between program and process . C01 05

1. b) Define system calls and explain their significance in an operating system. Discuss in detail how system calls act as an interface between user-level programs and the kernel. Illustrate the complete execution process with a suitable example. C01 05

OR

1. b) Describe the characteristics of a batch operating system in detail. Explain how jobs are processed in batches without user interaction and analyze its advantages and disadvantages compared to modern interactive operating systems.

2. a) A cloud computing platform is processing multiple tasks with varying burst times. The server uses Shortest Remaining Time First (SRTF) scheduling to ensure that tasks with the shortest remaining burst time are processed first. The tasks are submitted atdifferent times, and the platform needs to decide how to schedule them. Task Arrival Time (ms) Burst Time (ms).Once selected, each process is given the CPU for a fixed time quantum of 2 units. C02 05

Process	**Arrival Time**	**Burst Time**	**Priority**
P1	3	1	2
P2	1	4	1
P3	4	2	3
P4	0	6	2
P5	2	3	5

Simulate the execution of these tasks using SRTF scheduling.

i) Draw the Gantt chart.
ii)Calculate the waiting time and turnaround time for each task.

2. b) Consider a set of processes P1, P2, P3 and P4 with the following arrival times, burst times, and priorities: C02 05

Process	Arrival Time	Burst Time	Priority
P1	0	10	2
P2	1	15	1
P3	2	13	3
P4	3	11	2

The system uses a hybrid scheduling (RR+Priority) approach where processes are selected based on Priority Scheduling. Once selected, each process is given the CPU for a fixed time quantum of 5 units. After using its time quantum, if the process is not finished, it is placed back in the ready queue based on priority.
i) Draw the Gantt chart for the execution of processes.
ii) Find the average waiting time.

3. a) Semaphores are an essential synchronization tool used by operating systems to solve the critical section problem and manage concurrent access to shared resources. Describe the structure and operation of semaphores and evaluate their strengths and weaknesses as a synchronization mechanism. CO3 05

3. b) Explain the structure and working mechanism of a mutex lock with suitable diagrams or pseudocode.Discuss how locking and unlocking operations are implemented in an operating system. CO3 05

OR

3. b) A drone system schedules the following periodic tasks using Earliest Deadline First (EDF):
T1 (Obstacle Detection): Execution = 2 ms, Period = 6 ms, Deadline = 6 ms
T2 (Flight Stabilization): Execution = 1 ms, Period = 4 ms, Deadline = 4 ms
T3 (Video Streaming): Execution = 3 ms, Period = 12 ms, Deadline = 10 ms
Tasks:
i) Identify if any deadlines are missed under normal conditions.
ii) Suggest a modification to improve performance.